1 Questions repeated/similar to October 2025 & January 2026

(Rx. = Repeated question x)

R1. Evaluate

$$\int_1^4 \frac{15x^{3/2} + 15x^{1/2} - 30x^{-1/2}}{x+2}\, dx.$$

A. 20

B. 30

C. 40

D. 50

E. 60

F. 80

R2. A circle has equation

$$(x-p)^2+(y-q)^2=5.$$

The circle lies entirely in the region $y \leq 10-2x$. Find the maximum possible value of $2p+q$.

A. 5

B. $5-2\sqrt{5}$

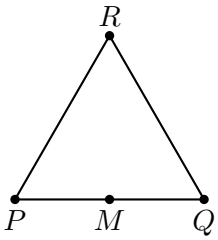
C. $5+2\sqrt{5}$

D. $10-2\sqrt{5}$

E. $10+2\sqrt{5}$

F. $15-2\sqrt{5}$

R3. PQR is an equilateral triangle of side length 2. Point M is the midpoint of PQ.
An ant starts at P and walks to M along PQ at constant speed. At the same time, another ant starts at Q and walks to R along QR at constant speed. They arrive at M and R at the same time.
What is the smallest distance between the two ants during their journeys?



A. $\frac{2\sqrt{21}}{7}$

B. $\sqrt{3}$

C. $\frac{\sqrt{7}}{2}$

D. $\frac{2\sqrt{3}}{7}$

E. $\frac{2\sqrt{7}}{7}$

F. $\sqrt{21}$

R4. Let $t_1, t_2, t_3, \ldots$ be a sequence of integers in arithmetic progression with common difference d. Let $S_n = t_1 + t_2 + \cdots + t_n$.
Suppose that there exists a positive integer k such that $t_k > t_{k+1}$. Which of the following statements must also be true?

I. $t_n < 0$ for all $n > k + 1$.

II. There exists a positive integer n such that $S_n < 0$.

III. $d < 0$.

A. None

B. I only

C. II only

D. III only

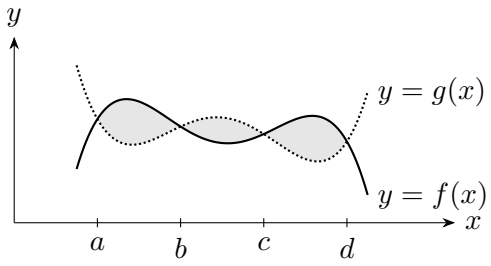
E. I and II only

F. I and III only

G. II and III only

H. I, II and III

R5. The graphs of two polynomials $y = f(x)$ (solid) and $y = g(x)$ (dotted) intersect at $x = a$, $x = b$, $x = c$ and $x = d$ only, as shown below.
Which of these expressions gives the total area enclosed between the two curves?



A. $\displaystyle\int_a^d (f(x) - g(x))\,dx - \int_b^c (f(x) - g(x))\,dx$

B. $\displaystyle\int_a^d (g(x) - f(x))\,dx + \int_b^c (f(x) - g(x))\,dx$

C. $\displaystyle\int_a^d (g(x) - f(x))\,dx - \int_b^c (f(x) - g(x))\,dx$

D. $\displaystyle\int_a^d (f(x) - g(x))\,dx + \int_b^c (g(x) - f(x))\,dx$

E. $\displaystyle\int_a^d (g(x) - f(x))\,dx + 2\int_b^c (g(x) - f(x))\,dx$

F. $\displaystyle\int_a^d (f(x) - g(x))\,dx + 2\int_b^c (g(x) - f(x))\,dx$

G. $\displaystyle\int_d^a (g(x) - f(x))\,dx + 2\int_b^c (f(x) - g(x))\,dx$

H. $\displaystyle\int_a^d (f(x) + g(x))\,dx - 2\int_b^c (f(x) + g(x))\,dx$

R6. $PQRS$ is a quadrilateral such that $PQ = QR$ and $RS = SP$.
Which of the following conditions, if any, is sufficient to guarantee that $PQRS$ is a rhombus?

A. PR bisects QS

B. QS bisects PR

C. PR is perpendicular to QS

D. $\angle QPS = \angle QRS$

E. $\angle PQR = \angle QRS$

F. None of these conditions

R7. A student attempts to solve

$$(\log_2 x)^2 - 5\log_2 x + 6 = 0, \qquad x > 0.$$

The student's working is as follows:

I: $2\log_2 x - 5\log_2 x + 6 = 0,$
II: $-3\log_2 x + 6 = 0,$
III: $\log_2 x = 2,$
IV: The only solution of the original equation is $x = 4$.

Which of the following best describes this attempt?

A. The method is valid and 4 is the only solution.

B. The method is invalid and the solutions are 2 and 4.

C. The method is invalid and the solutions are 4 and 8.

D. The method is invalid and the solutions are 4 and 16.

E. The method is invalid and the solutions are 2 and 8.

R8. The first three terms of an arithmetic sequence are

$$4x - 9, \qquad x - 7, \qquad 2x - 11.$$

Find the fourth term of the sequence.

A. $-\frac{19}{2}$

B. $-\frac{21}{2}$

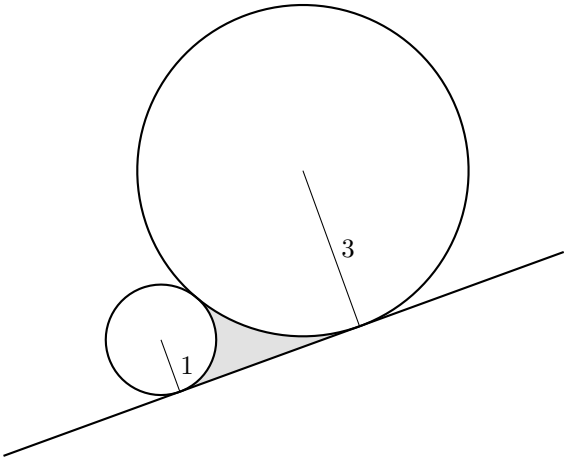
C. $-\frac{23}{2}$

D. -11

E. $-\frac{17}{2}$

F. $-\frac{25}{2}$

R9. Two circles have radii 1 and 3. They touch the same straight line and also touch each other externally.
Find the area of the shaded region enclosed between the two circles and the line.



A. $4\sqrt{3} - \dfrac{19\pi}{6}$

B. $4\sqrt{3} - \dfrac{11\pi}{6}$

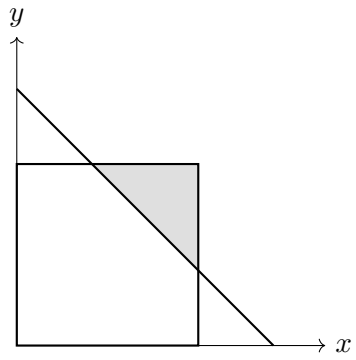
C. $4\sqrt{3} - \dfrac{13\pi}{6}$

D. $4\sqrt{3} - \dfrac{17\pi}{6}$

E. $4\sqrt{3} - \dfrac{7\pi}{6}$

F. $4\sqrt{3} + \dfrac{11\pi}{6}$

R10. A square of area 1 is placed with one corner at the origin and its sides along the positive coordinate axes. A straight line of gradient -1 forms a right-angled triangle with the coordinate axes, and the area of this triangle is 1. The line intersects the top and right sides of the square, forming the shaded region shown.
Find the area of the shaded region.



A. $3 - 2\sqrt{2}$

B. $2\sqrt{2} - 2$

C. $3 - \sqrt{2}$

D. $\sqrt{2} - 1$

E. $2 - \sqrt{2}$

F. $\dfrac{3 - 2\sqrt{2}}{2}$

R11. Let

$$f(x) = x^2 - 5x + 4.$$

The graphs of $y = f(|x|)$ and $y = -x + k$, where k is a real constant, intersect at exactly two distinct points.
Find the complete set of possible values of k.

A. $-5 < k < 0$ or $k > 4$

B. $k > 4$

C. $-5 < k < 0$

D. $k < -5$

E. $0 < k < 4$

F. $k = 0$ or $k = 4$

R12. The first three terms of an arithmetic progression are

$$\log_b 4, \qquad x, \qquad \log_b 16,$$

where $b > 0$ and $b \neq 1$. Find x.

A. $\log_b 8$

B. $\log_b 10$

C. $\frac{1}{2} \log_b 20$

D. $\sqrt{2} \log_b 4$

E. $\frac{1}{2} \log_b 12$

F. $\frac{2}{3} \log_b 16$

R13. Let $p > 0$. Consider

S: $\cos x = -\sin(x - p)$ for every real x if p is an odd multiple of $\pi/2$.

T: $\cos x = -\cos(x - p)$ for every real x only if p is an odd multiple of π.

Which of S, the converse of S, and T are true?

R14. The polynomial

$$4x^4 + 16x^3 + mx^2 - 16x + n$$

can be written as $(2x^2 + px + q)^2$, where m, n, p, q are real constants. Find $m + 2n$.

A. 0

B. 16

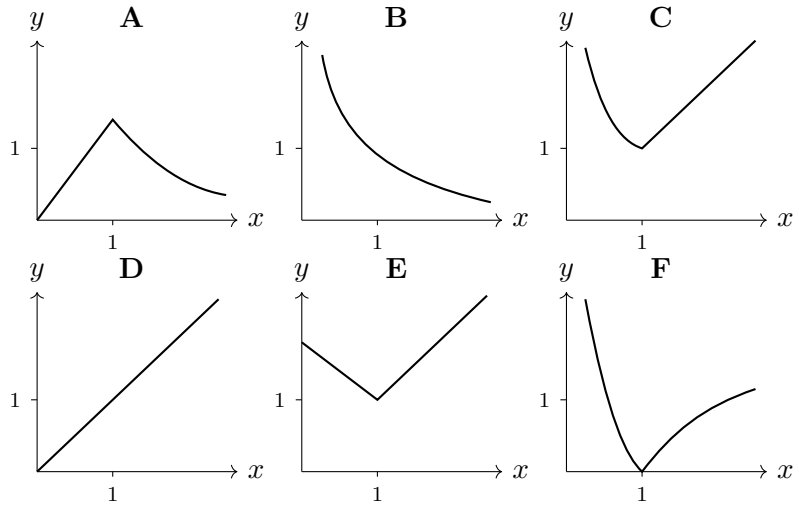
C. 24

D. 32

E. 68

R15. Which of the following is the graph of

$$y = 2^{|\log_2 x|} \ ?$$



R16. Let a, b, c, d be real, with $b \neq 0$ and $d \neq 0$. Consider

$$R: \ \frac{a}{b} < -\frac{c}{d}, \qquad S: \ ad + bc < 0.$$

Which of the following describes the relationship between S and R?

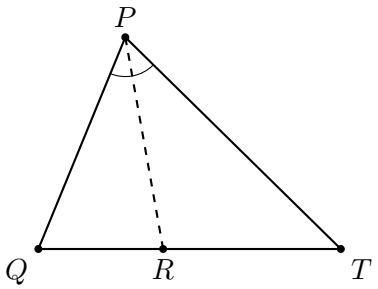
A. S is necessary and sufficient for R.

B. S is sufficient but not necessary for R.

C. S is necessary but not sufficient for R.

D. S is neither necessary nor sufficient for R.

R17. In acute triangle PQT, the point R lies on QT and PR bisects $\angle QPT$. Express QR/RT in terms of the angles at Q and T, choosing from sine and tangent ratios.



R18. The cubic $f(x) = ax^3 + ax^2 + 3ax + b$, where $a \neq 0$ and $b \neq 0$, has the root $x = -b$. Find the range of possible values of a.

A. $0 < a \leq \frac{4}{11}$

B. $0 < a < \frac{4}{11}$

C. $0 < a \leq \frac{1}{3}$

D. $0 < a < \frac{1}{3}$

E. $a \geq \frac{4}{11}$

F. $a \leq 0$

G. $-\frac{4}{11} \leq a < 0$

H. $a \geq \frac{1}{3}$

R19. The interior angles of a triangle satisfy $0 < A < B < C < 180°$. Which of the following necessarily hold?

I. $\sin A < \sin B < \sin C$.

II. $\cos A > \cos B > \cos C$.

III. All three tangents are defined and $\tan A < \tan B < \tan C$.

A. None

B. I only

C. II only

D. III only

E. I and II only

F. I and III only

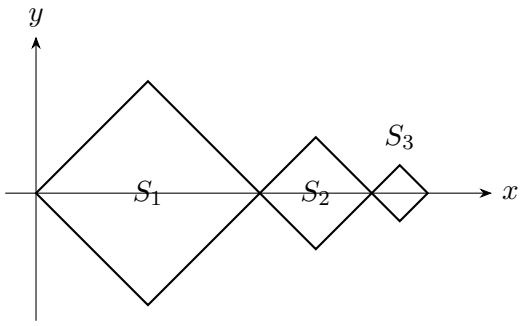
G. II and III only

H. I, II and III

R20. An infinite sequence of squares $S_1, S_2, \ldots$ is arranged along the x-axis as shown. The x-coordinate of the leftmost vertex of S_n is x_n, where

$$x_{n+1} = x_n + 2^{3-n}, \qquad x_1 = 0.$$

The rightmost vertex of S_n is the leftmost vertex of S_{n+1}.
What is the sum of the areas of all the squares?



A. $\frac{8}{3}$

B. $\frac{32}{3}$

C. $\frac{64}{3}$

D. 8

E. 16

R21. Given $0 < x < 2\pi$, find the complete set of values of x for which $\cos x$, $\cos 2x$ and $\cos 4x$ are all negative.

A. $\frac{\pi}{8} < x < \frac{3\pi}{8}$ or $\frac{13\pi}{8} < x < \frac{15\pi}{8}$

B. $\frac{\pi}{2} < x < \frac{3\pi}{4}$ or $\frac{5\pi}{4} < x < \frac{3\pi}{2}$

C. $\frac{5\pi}{8} < x < \frac{7\pi}{8}$ or $\frac{9\pi}{8} < x < \frac{11\pi}{8}$

D. $\frac{5\pi}{8} < x < \frac{3\pi}{4}$ or $\frac{5\pi}{4} < x < \frac{11\pi}{8}$

E. $\frac{\pi}{4} < x < \frac{3\pi}{8}$ or $\frac{5\pi}{8} < x < \frac{3\pi}{4}$ or $\frac{5\pi}{4} < x < \frac{11\pi}{8}$ or $\frac{13\pi}{8} < x < \frac{7\pi}{4}$

R22. Consider the simultaneous equations

$$y = 2^k x, \qquad y = x + k,$$

where k is real. Which statement is true?

A. The equations have no solution (x, y) for any value of k.

B. The equations have no solution for one value of k, and a unique solution for every other value.

C. The equations have no solution for one value of k, and infinitely many solutions for every other value.

D. The equations have a unique solution for every value of k.

E. The equations have a unique solution for one value of k, and no solution for every other value.

F. The equations have a unique solution for one value of k, and infinitely many solutions for every other value.

G. The equations have infinitely many solutions for every value of k.

H. The equations have infinitely many solutions for one value of k, and no solution for every other value.

I. The equations have infinitely many solutions for one value of k, and a unique solution for every other value.

R23. Six positive integers are arranged in ascending order to form a data set. The mean of the data set is 6, the median is 4.5, the range is 10, and the unique mode is 4.
Which of the following statements must be true?

I. There exist exactly 3 possible data sets with the given information.

II. The number 4 appears exactly 2 times in the data set.

III. The difference between the 5th value and the 2nd value is always odd.

A. None of the statements

B. I only

C. II only

D. III only

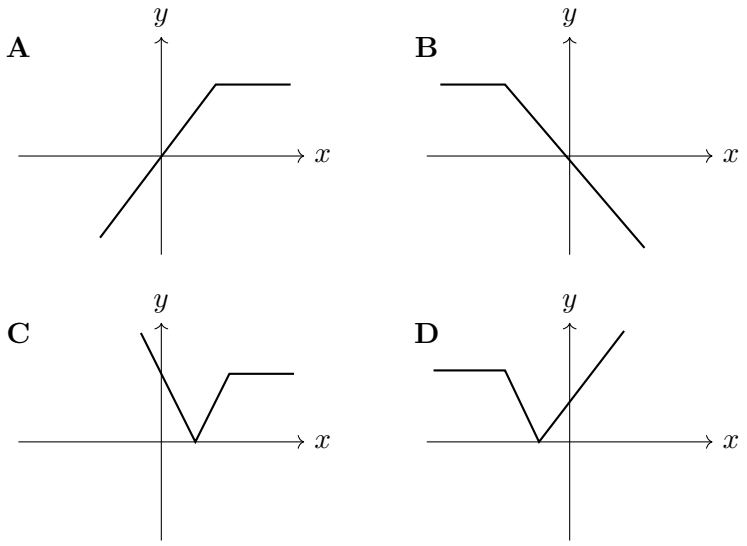
E. I and II only

F. I and III only

G. II and III only

H. I, II and III

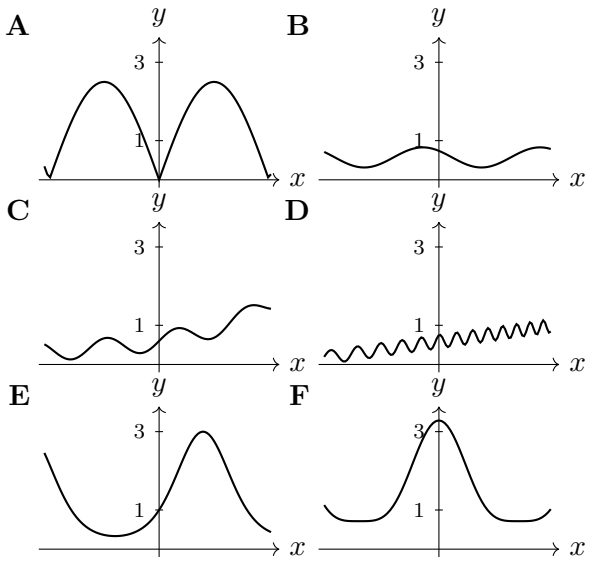
R24. Which of the four graphs below represents $y = \big|x - |x-1|\big|$?



R25. Sketch the function

$$y = 3^{\sin x}.$$

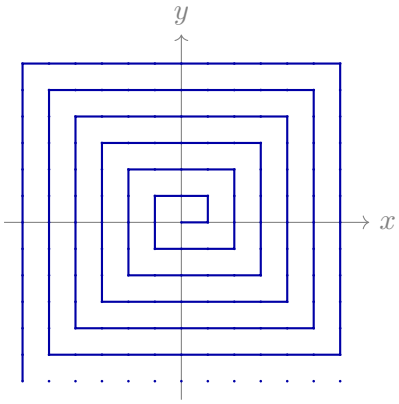
In particular, determine its period, range, value and direction of change at $x = 0$. Which of the graphs has these properties?



R26. How many points of intersection are there between the graphs of

$$y = 2^{x^3-3x+2} \qquad \text{and} \qquad y = x\,?$$

R27. A square spiral is drawn on the integer-coordinate grid as shown. It passes through $(0, 0)$ and successive integer grid points, turning at the corners of progressively larger squares. The length of the spiral up to $(-1, -1)$ is 6, following the path shown in the diagram. What is the length of the spiral up to $(8, 10)$?



A. 108

B. 168

C. 270

D. 274

E. 380

F. 382

G. 456

R28. Find the value of

$$\sum_{n=1}^{10} \sqrt{1 + 8 \sum_{r=1}^{n} r}\,.$$

A. 110

B. 115

C. 120

D. 121

E. 125

F. 130

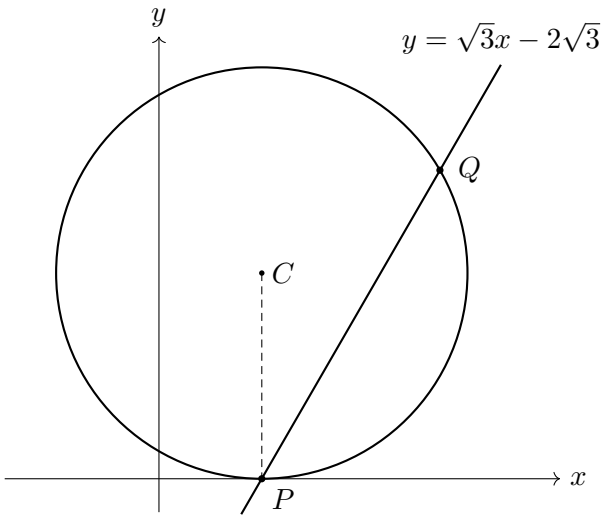
G. 55

H. 210

2 Unique October 2026 questions (afaik)

(Ux. = Unique question x)

U1. A circle of radius 4 has its centre above the x-axis and is tangent to the x-axis at $P = (2, 0)$. The line $y = \sqrt{3}x - 2\sqrt{3}$ meets the circle again at Q. Find the length of the minor arc PQ.



A. $\frac{\pi}{3}$

B. $\frac{2\pi}{3}$

C. $\frac{4\pi}{3}$

D. $\frac{8\pi}{3}$

E. $\frac{16\pi}{3}$

U2. How many distinct real solutions does

$$4^{1+\sin^2\theta} + 4^{\frac{1}{2}+\cos^2\theta} = 12$$

have for $0 \leq \theta \leq 2\pi$?

A. 0

B. 1

C. 3

D. 5

E. 7

F. 8

U3. The circle $(x-6)^2+(y-4)^2=9$ is reflected in the line $y+5x=p$. Its image has equation $x^2+y^2+8x-4y+11=0$. Find p.

U4. The cubic $y = qx^3 + rx^2 + sx + t$ has turning points at $x = -2$ and $x = \frac{2}{3}$ and passes through $(0, 7)$ and $(1, 5)$. Find the value of y when $x = -1$.

U5. The line $x + 3y = 4$ meets the curve $x^2 + 3xy + 2y^2 = 6$ at P and Q. Find the distance PQ.

U6. A biased six-sided die is numbered 1 to 6. It is known that

$$P(1) = 0.2, \qquad P(\text{even}) = 0.4, \qquad P(\text{prime}) = 0.7.$$

Find $P(2)$ or state that there is insufficient information.

U7. Find the number of distinct real solutions of

$$|x^3 + 3x^2 - 9x| = 5.$$

A. 0

B. 1

C. 2

D. 3

E. 4

F. 5

G. 6

U8. Find the maximum possible value of

$$f(x) = 3\cos^2 x + 3\sin x + 4, \qquad x \in \mathbb{R}.$$

U9. The graph of $y = x^3 - \frac{5}{4}x$ is translated by the vector $\binom{k}{k}$. Find all real k for which the translated graph intersects the original graph.

U10. The polynomial f has precisely four distinct real roots $a < b < c < d$. It is given that

$$\int_a^c f(x)\,dx = 3, \qquad \int_b^d f(x)\,dx = 5, \qquad \int_a^b f(x)\,dx = 3\int_c^d f(x)\,dx.$$

Find the area of the largest region bounded by $y = f(x)$ and the x-axis.

U11. For $0 < x < \pi/2$, simplify

$$\log_{10}\bigl(\cos x + \tan x \sin x\bigr).$$

U12. Find $(1.002)^7 - (0.998)^7$ correct to eight decimal places.

U13. Let $f(x) = px^2 + qx + r$ where $p \neq 0$. The line $y = f'(x)$ is tangent to the parabola $y = f(x)$ and has the same y-intercept as that parabola. Find the x-coordinate of the turning point of f.

U14. Let S be the sum of every integer from 8 to 8^n inclusive, where $n > 1$ is an integer. If 2^p is the highest power of 2 dividing S, find p.

U15. The function $f(x) = x^3 + 5x^2 + 3x + c$ has the property that f and f' share a linear factor. Find the sum of all possible values of c.

U16. Find all real x satisfying

$$\log_4\left(\frac{1}{x}\right) + 3\log_4(x^2) + (2\log_4 x)^2 = 6.$$

U17. Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(0) = 0$, and let a, b, c, d be nonzero constants. The curves

$$y = af(x + b), \qquad y = f(cx) + d$$

intersect both on the x-axis and on the y-axis. Which condition is sufficient?

A. $af(b) = d$ and $f(-bc) = d$.

B. $af(b) = d$ and $f(-bc) = -d$.

C. $af(b) = -d$ and $f(-bc) = d$.

D. $af(b) = -d$ and $f(-bc) = -d$.

U18. An arithmetic progression has n terms, first term a and difference d, with n, a, d positive integers. Knowing only whether each of n, a, d is even or odd, can one determine the parity of the sum?

I. This information always suffices.

II. It need not suffice, but it will if n is also known exactly.

III. It need not suffice, but it will if a is also known exactly.

IV. It need not suffice, but it will if d is also known exactly.

Which statement is correct?

U19. A teacher writes $x^2 + bx + c = 0$. The first student copies b correctly but c incorrectly and finds roots -4 and 8. The second copies c correctly but b incorrectly and finds roots -3 and 4. Find the roots of the teacher's equation.

U20. Which of the following numbers is prime?

A. $79^2 - 70^2$.

B. $19^7 - 1$.

C. $11! - 7$.

D. $2^{13} - 1$.

E. $6^{15} - 1$.

U21. Suppose k is real and

$$\int_0^k \sin x \, dx = 0.$$

Which statements necessarily follow?

I. $k = 2\pi$.

II. $\displaystyle\int_0^k \sin 2x \, dx = 0.$

III. $\displaystyle\int_0^k \sin^2 x \, dx = 0.$

U22. The quantities P, Q, R, S are given as towers of the numbers 2 and 3:

$$P = 2^{2^{2^2}}, \quad Q = 2^{3^3}, \quad R = 3^{2^{2^2}}, \quad S = 3^{2^3}.$$

Exponentiation is evaluated from the top down, so that a^{b^c} means $a^{(b^c)}$. Arrange the four quantities in increasing order.

U23. Let $b, c, d, e \in \mathbb{R}$, and suppose $x^2 + bx + c = (x + d)^2 + e$ for all x. Set

$$P: \ b > 0 \text{ or } c > 0, \qquad Q: \ d > 0 \text{ or } e > 0.$$

Is P necessary, sufficient, both or neither for Q?

U24. Consider the claim: "If a quadratic f has no real roots, then $f(N)$ is prime for every positive integer N." Which of the following are counterexamples?

I. $f(x) = x^2 + x + 1$ and $N = 4$.

II. $f(x) = x^2 + 1$ and $N = 2$.

III. $f(x) = x^2 - 1$ and $N = 3$.

U25. For $x \geq 0$, write $\{x\} = x - \lfloor x \rfloor$ for the fractional part of x. Evaluate

$$\int_0^5 \big(2\{x\} - x\big)\, dx.$$

U26. Let a, b be positive integers. Which of the following are always true?

I. a^b is a square if and only if b is even.

II. $a + b$ is a square if both a and b are squares.

III. ab is a square only if at least one of a, b is a square.

U27. Let m, n be positive integers with $m < n$. Set

$$P: \ m \text{ has two digits and } n \text{ has three digits}, \qquad Q: \ mn \text{ has four digits.}$$

Is P necessary and/or sufficient for Q?

U28. A student writes

$$\begin{aligned}
\text{I: } & \cos^2\theta = 1 - \sin^2\theta, \\
\text{II: } & (\cos^2\theta)^{3/2} = (1 - \sin^2\theta)^{3/2}, \\
\text{III: } & \cos^3\theta = (1 - \sin^2\theta)^{3/2}, \\
\text{IV: } & \cos^3\theta + 3 = (1 - \sin^2\theta)^{3/2} + 3, \\
\text{V: } & (\cos^3\theta + 3)^2 = \left((1 - \sin^2\theta)^{3/2} + 3\right)^2.
\end{aligned}$$

Then $\theta = \pi$ is substituted and an apparent contradiction is obtained. Where is the first error?

U29. Let $a = 2^p 3^q$ and $b = 2^x 3^y$, where p, q, x, y are nonnegative integers. Their highest common factor is even, and their lowest common multiple is a perfect square. Which statements must be true?

I. $px > 0$.

II. Both p and x are even.

III. qy is even.

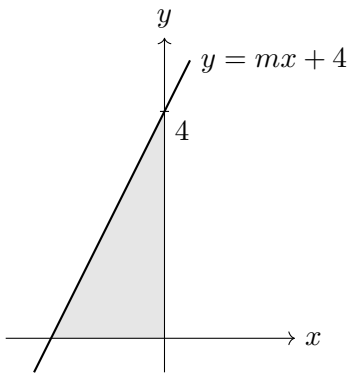
U30. The line $y = mx + 2$ is tangent to the parabola $y = x^2 + px + q$, for real m, p, q. Which statements must hold?

I. $q \geq 2$.

II. If $q = 2$, then $p = m$.

III. If $m = 0$, then $p = 0$.

U31. The gradient of $y = mx + 4$ depends on time $t > 0$ according to $m = 2\sqrt{t}$. Let P be the area enclosed by the line and the coordinate axes. At what time is P decreasing at one square unit per second?



U32. Given $\int_a^b f(x)\,dx = R$, where $0 < a < b$ and $R \in \mathbb{R}$, which of the following must also equal R?

A. $\int_a^b -f(x)\,dx.$

B. $\int_{-a}^{-b} -f(x)\,dx.$

C. $\int_{-b}^{-a} -f(x)\,dx.$

D. $\int_a^b f(-x)\,dx.$

E. $\int_{-a}^{b} f(-x)\,dx.$

F. $\int_{-b}^{-a} f(-x)\,dx.$

U33. The line L is normal to $y = -4/x^2$ at $x = \sqrt{2}$. It meets the x-axis at P and the y-axis at Q. Find the midpoint of PQ.

U34. The trapezium rule is used to estimate the area under each of the following curves from $x = 2$ to $x = 2.5$:

I. $f(x) = |\cos x|$ (radians).

II. $g(x) = -\tan x$ (radians).

III. $h(x) = |\log_{10}(x/3)|$.

For which does the trapezium rule give an underestimate?

U35. Let $f(x) = x^2 - 6x + 9$. How many distinct real solutions does the equation

$$f(f(f(x))) = 0$$

have?

U36. Find the sum of the real solutions of

$$9^x - 3^{x+2} + 20 = 0.$$

Express it using logarithms to base 10.

U37. Let $x \in \mathbb{R}$. Consider

$$P: \ x^{12} = 16, \qquad Q: \ x^{15} = 32.$$

Is P necessary and/or sufficient for Q?

U38. Let f be a real polynomial and $g(x) = [f(x)]^2$. Which statement is *not necessarily true*?

A. If f has a local minimum at $x = a$, then g has a stationary point there.

B. If f has a local maximum at $x = a$, then g has a stationary point there.

C. If g has a local minimum at $x = a$, then f has a stationary point there.

D. If g has a local maximum at $x = a$, then f has a stationary point there.

U39. Consider the statement: "Every integer greater than 1 can be written as $p + a^2$, where p is prime and a is an integer." Which of the integers $3, 4, 5, 6, 7, 8, 9, 10$ is a counterexample?

U40. Let $p(x) = ax^3 + bx^2 + cx + d$, where $a \neq 0$. Is $ad < 0$ necessary, sufficient, both, or neither for p to have at least one positive real root?

U41. Three cards have initial left-to-right order X, Y, Z. On odd-numbered moves, swap the left and middle cards; on even-numbered moves, swap the middle and right cards. What is their order after exactly 200 swaps?

U42. A student gives this proof that, for primes $p, q > 3$, the integer $p^2 - q^2$ is divisible by 3:

I. $p^2 - q^2$ is divisible by 3 if and only if at least one of $p + q$ and $p - q$ is divisible by 3.

II. Every prime $p > 3$ is of the form $3k + 1$ or $3k - 1$ for some integer k.

III. Consequently, one of $p + q$ and $p - q$ is divisible by 3.

IV. Hence 3 divides $p^2 - q^2$.

Which step, if any, contains the first mathematical error?

U43. Suppose $x^2 + px + q = 0$ has two real roots $x_1 < x_2$. Keeping q fixed, increase p so the new equation has two distinct real roots $z_1 < z_2$. Which inequalities between $z_1 + z_2$, $x_1 + x_2$, $z_2 - z_1$ and $x_2 - x_1$ are possible?

U44. Let f be a polynomial and define $F(x) = \int_0^x f(t)\,dt$ for $x > 0$. Consider

$$P:\ f'(x) > 0 \text{ for all } x > 0, \qquad Q:\ F'(x) \geq 0 \text{ for all } x > 0.$$

Which of the following are true?

I. P is necessary for Q.

II. Q is necessary for P.

III. If both P and Q hold, then $F'(x) > 0$ for all $x > 0$.

U45. A real geometric series $x_1 + x_2 + \cdots$ converges to 1. The odd-indexed series $x_1 + x_3 + x_5 + \cdots$ has sum S. Find the complete range of S. (The common ratio may be zero.)

U46. Someone owns at least ten pens, each of which is blue, green or red. In every selection of ten pens, at least one is blue, at least two are green and at least three are red. What is the largest possible number of pens owned?

U47. Let $f(x) = \sin^2 x - 4\sin x + 5$. A student argues:

I. On putting $u = \sin x$, one obtains $f(x) = u^2 - 4u + 5 = (u-2)^2 + 1$.

II. This quadratic in u has its minimum over $\mathbb{R}$ at $u = 2$.

III. Since $-1 \le \sin x \le 1$, this minimum is not attained.

IV. Therefore $y = f(x)$ has no turning points.

Which is the first invalid conclusion, if any?

U48. For which real values of k does

$$\sin x = \cos(k - x)$$

hold for every real x? Give your answer using an integer parameter.

U49. The statement

$$\text{“For every } k > 0, \quad \int_{-k}^{k} f(x)\,dx > 0\text{”}$$

is false. Write the precise negation of this statement.

U50. The side lengths $a_1, a_2, \ldots$ of a sequence of squares satisfy

$$a_1 = 1, \qquad a_2 = \frac{1}{2}, \qquad a_n = \frac{a_{n-1}^2}{a_{n-2}} \quad (n \geq 3).$$

Find the sum of their perimeters.

U51. Three distinct integers are selected from $1, 2, \ldots, 10$ without regard to order. How many selections have exactly one even number and exactly one multiple of 3?

U52. Two independent fair dice are rolled. Let M be the score on the first die and S be the sum of the scores. For how many ordered pairs (m, s), with $1 \leq m \leq 6$ and $2 \leq s \leq 12$, are the events $\{M = m\}$ and $\{S = s\}$ independent?

U53. For real $x > -2$, consider

$$P:\ \log_2(x+2) > 2, \qquad Q:\ x > 5.$$

Is Q necessary and/or sufficient for P?

U54. Let $0 < x < \pi$, with $x \neq \pi/2$. Which strict orderings of $\sin x$, $\cos x$ and $\tan x$ are impossible?

U55. Find all positive real solutions of

$$(\log_3 x)^2 - \log_3(x^2) = 0.$$

In one version, find the sum of all the solutions.

U56. A differentiable function satisfies

$$f'(x) = (x-1)(x-2)(x-4).$$

The equations $f(x) = a$ and $f(x) = b$, where $a \neq b$, each have exactly three distinct real solutions. How many distinct solutions can

$$f(x) = \frac{a+b}{2}$$

have?

U57. Determine the difference between the two smallest nonnegative solutions (in degrees) of

$$16\cos^2 x - 3\tan^2 x = 11.$$

U58. Find the sum of all solutions in $-\pi/2 < x < 3\pi/2$ of

$$\sqrt{3}\tan^3 x - 2\tan^2 x - \sqrt{3}\tan x = 0.$$

U59. Eight people A, B, C, D, E, F, G, H sit equally spaced around a round table, in that order. Each neighbouring pair independently shakes hands with probability $1/2$. What is the probability that A and E are connected by a path of handshakes along the table?

U60. Let n straight lines be drawn in a plane such that no three lines are concurrent. Suppose k of these lines are mutually parallel. The remaining $n-k$ lines are all non-parallel to one another and to the first k lines.
If $n = 10$ and there are exactly 35 distinct points of intersection, find k.

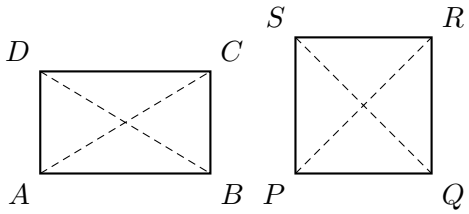
U61. Let $f(x) = 2^{-x}$ and let p be real. Find the condition on p necessary and sufficient for the equation $f(x) = p$ to have exactly one solution with $x > 4$.

U62. Let a, m, n be positive integers. Consider

$$P: \ m \text{ is a multiple of } n, \qquad Q: \ \gcd(a, m) > 1 \text{ whenever } \gcd(a, n) > 1.$$

Determine whether P is necessary and/or sufficient for Q.

U63. Two rectangles $ABCD$ and $PQRS$ have diagonal lengths satisfying $AC = PR$ and $BD = QS$. Does this information suffice to conclude that the two rectangles have equal areas? Is the condition necessary, sufficient, both or neither?



U64. Let f be a cubic polynomial. Consider

A: $f(x)$ has at least one strictly positive real root and at least one strictly negative real root.

B: $f(-x)$ has at least one strictly positive real root and at least one strictly negative real root.

Is B necessary and/or sufficient for A?

U65. Sketch

$$y = e^{\sin x},$$

indicating its range, period, intercept on the vertical axis and location of maxima and minima.

U66. The polynomial

$$f(x) = x^4 + ax^3 + bx^2 - x + 30$$

has the factors $(x - 5)$ and $(x - 3)$. Find the sum of its coefficients.

U67. Find the sum of all real solutions of

$$\sin^2 x - \cos x = \cos^2 x$$

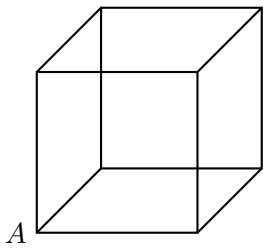
for $-\pi/2 \leq x \leq 3\pi/2$.

U68. The lengths of the three sides of a right-angled triangle, in increasing order, are $a, a+d, a+2d$, where $a, d > 0$. Express a in terms of d.

U69. Evaluate

$$\sum_{n=1}^{\infty} \left(3^{1-2n} + 2^{-2n}\right).$$

U70. An ant starts at vertex A of a cube. Each minute it travels along one incident edge to an adjacent vertex, independently choosing between the three edges with equal probability. Find the probability it is back at A exactly four minutes later.

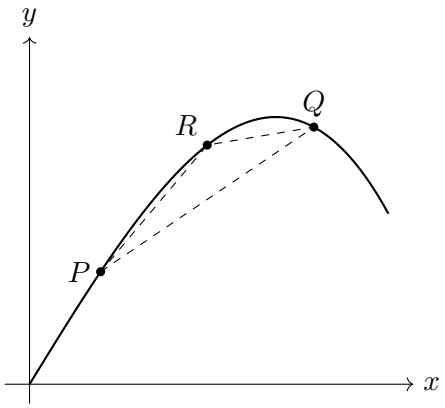


U71. The first three terms of an arithmetic progression add to 15. If 8 is added to the first term, 1 to the second, and 2 to the third, the three resulting numbers form a convergent geometric progression. Find the sum to infinity of that geometric progression.

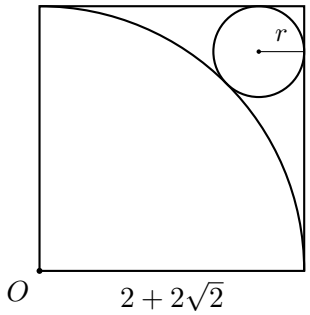
U72. The curve C is given by

$$y = 36x - x^3.$$

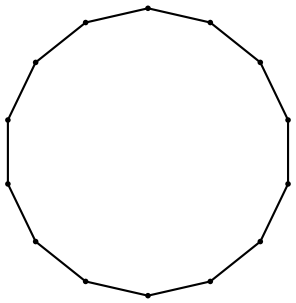
Let $P = (1, 35)$, $Q = (4, 80)$ and $R = (a, b)$ be three points on C with $1 < a < 4$. Find the coordinates of R which maximise the area of triangle PQR.



U73. A quarter-circle of radius $2 + 2\sqrt{2}$ and a full circle lie inside a square of side $2 + 2\sqrt{2}$. The quarter-circle is centred at one corner of the square. The full circle is tangent to the quarter-circle and to the two sides of the square that do not meet at that corner. Find the radius of the full circle.



U74. The vertices of a regular 14-gon are labelled. How many non-congruent isosceles triangles can be formed by choosing three of its vertices?



Regular 14-gon (schematic)

U75. Let $x_1, \ldots, x_{10}$ be ten real numbers. Set P: more than five of them are negative; Q: their mean is negative. Determine whether P is necessary or sufficient for Q.

U76. Let a, b, c be real, with $a \neq 0$ and $b > 0$. The equation $ax^2 + bx + c = 0$ has roots a and b. Which of the statements $a + c < 0$, $a + c = 0$, $a + c > 0$, or "$a + c$ can have any sign" must be true?

U77. Five distinct mathematicians and five distinct historians are to be arranged in a row. An arrangement is admissible when no three consecutive people belong to the same subject. Let N denote the number of admissible arrangements. Consider:

I. Exactly 84 admissible sequences of subject labels are possible, before choosing the identities.

II. In every admissible arrangement at least four of the nine adjacent pairs contain people from different subjects.

III. If both ends must be mathematicians, the number of arrangements is $24(5!)^2$.

IV. If the ten people sit in a circle with the same restriction (including the circular join), up to rotation but not reflection, there are $N/10$ arrangements.

Which are correct?

U78. Let $f(x) = (x-1)^2 + 1$, and define $f^{(n)}$ as the n-fold composition of f with itself. How many real solutions does

$$f^{(2026)}(x) = 2$$

have?

U79. An arithmetic progression has nonzero common difference d, first term a and sums S_n. Given $S_m = S_n$ for integers $1 \leq m < n$, which of the following must be true?

I. $ad < 0$.

II. If $m + n$ is odd, a term of the progression is zero.

III. $S_{m+n} = 0$.

U80. Independent fair six-sided dice give positive integers p and q. Find the probability that

$$\sin^2 x + p \sin x + q = 0$$

has at least one real solution x.

U81. Consider the claim "Every positive integer n that is not congruent to 1 modulo 6 is composite." Write its precise logical negation.

U82. A real geometric series $x_1 + x_2 + \cdots$ converges to 1 and has nonzero first term. The common ratio may be zero. Let

$$S = x_1^2 + x_2^2 + x_3^2 + \cdots .$$

Determine all possible values of S.

U83. Let

$$I_1 = \int_0^1 2^{x/2}\,dx, \qquad I_2 = \int_0^1 2^{x}\,dx, \qquad I_3 = \int_0^1 2^{\sqrt{x}}\,dx.$$

Arrange I_1, I_2, I_3 in increasing order.

U84. Find every real solution for which the expression is defined to

$$(x+6)^{x-3} = 1.$$

U85. For real x and a positive constant k, solve

$$k^{x^2} > k^x.$$

Distinguish the cases $k > 1$, $0 < k < 1$ and $k = 1$.

U86. For integers a, b, c, determine which of the following assertions are true:

I. If $a + b$ is even, then ab is even.

II. If ab is even, then $a + b$ is even.

III. If $a + b + c$ is even, then abc is even.

U87. Find the greatest real solution of

$$x - 6\sqrt[3]{x} - 4 = 0.$$

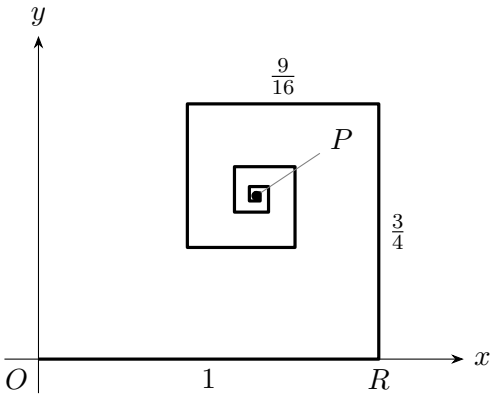
U88. An arithmetic sequence $t_1, t_2, \ldots$ has common difference d and partial sums S_n. For some positive integer k, $S_{k+1} < S_k$. Determine which conditions necessarily follow:

I. $t_n < 0$ for all $n \geq k + 1$.

II. There is a positive integer n for which $S_n < 0$.

III. $d < 0$.

U89. A spiral is formed by a sequence of straight-line segments. The first segment is OR, which has length 1. Each subsequent segment is perpendicular to the preceding segment and has length $\frac{3}{4}$ of the preceding segment. The segments spiral towards a limiting point P, as shown in the diagram.



Which of the following is the x-coordinate of P?

A. $\frac{16}{25}$

B. $\frac{12}{25}$

C. $\frac{4}{5}$

D. $\frac{28}{25}$

E. 4

U90. Find all x with $-\pi < x < \pi$ satisfying

$$4(\cos^2 x - \sin^2 x) - (2\sqrt{2} + 2)\cos x + (\sqrt{2} + 2) = 0.$$

U91. Let p, q, r be integers with $p > q$, and suppose

$$p^2 + q^2 = 2r^2.$$

Which of the following statements must be true?

I. p and q are both odd or both even.

II. The numbers $p - q$, $p + q$ and $2r$ can be the side lengths of a right-angled triangle.

III. r is the mean of p and q.

U92. Consider the infinite sequence of positive integers whose first three terms are

$$10001, \quad 100010001, \quad 1000100010001, \quad \ldots$$

where the nth term consists of $n+1$ digits equal to 1, with each pair of adjacent 1s separated by three zeros. Which statements must be true for this sequence?

1. Every term can be written as the sum of two square numbers.
2. No term in the sequence is a square number.
3. No term in the sequence is a prime number.

A. None of them.

B. (1) only.

C. (2) only.

D. (3) only.

E. (1) and (2) only.

F. (1) and (3) only.

G. (2) and (3) only.

H. (1), (2) and (3).

U93. Given

$$\sqrt{y} + \sqrt{y-1} = \left(\sqrt{x} + \sqrt{x-1}\right)^2,$$

find y in terms of x.

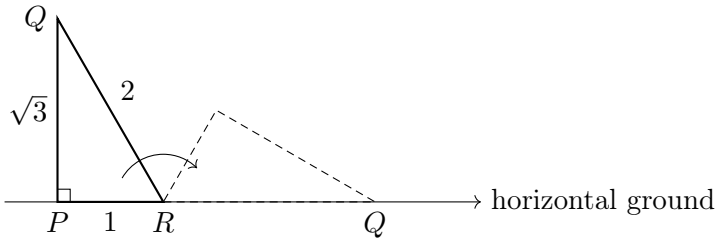
U94. How many real solutions does the equation

$$8^{\cos^2 x} + 4^{\sin x + 1} = 13$$

have for $0 < x < 2\pi$?

U95. A right-angled triangle PQR has $PR = 1$, $PQ = \sqrt{3}$ and $QR = 2$, with the right angle at P. It stands with PR on a horizontal line and Q vertically above P, as shown. The triangle is toppled to the right about R until QR lies on the line. It continues to be toppled to the right in this way, each time about its rightmost vertex on the line, until it is first back in its original orientation.

Find the total distance travelled by Q.



A. $\dfrac{11\pi}{6}$

B. $\dfrac{(4+3\sqrt{3})\pi}{6}$

C. $\dfrac{(8+3\sqrt{3})\pi}{6}$

D. $\dfrac{(4+5\sqrt{3})\pi}{6}$

E. $\dfrac{(8+3\sqrt{3})\pi}{2}$

F. $3+\sqrt{3}$

U96. Let n be an integer with $n \geq 2$. Consider the statement

$$(*) \qquad \text{For every integer } a, \quad n \mid a^2 \implies n \mid a.$$

Which of the following statements about $(*)$ is true?

A. $(*)$ is true for all integers $n \geq 2$.

B. $(*)$ is false for all integers $n \geq 2$.

C. $(*)$ is true if n is prime.

D. $(*)$ is true only if n is prime.

E. $(*)$ is true if and only if n is prime.

U97. Evaluate

$$\sqrt{10} + \sqrt{100} + \sqrt{1000} + \sqrt{10000} + \cdots + \sqrt{1\,000\,000\,000}.$$

U98. The polynomial $f(x) = ax^2 + bx + c$ leaves remainder b upon division by x, and remainder a upon division by $x + 2$.
Which of the following is true?
(I think this question was phrased like 'what is the value of $a + c$' but it should still test the same idea like this)

A. $a + c$ must equal 0.

B. $a + c$ must equal 4.

C. $a + c$ must be positive.

D. $a + c$ must be negative.

E. $a + c$ can take any real value.

3 Incomplete/unverified questions

These questions either don't provide enough information or haven't been corroborated by more than 1 individual. I have kept them so the material isn't lost, but will try to figure out what the correct phrasing/provenance is at some point.

(Xx. = Incomplete/unverified question x)

Probably should not have used X ...

X1. Suppose a, b, c are integers and $a \neq 0$. Which of the conditions below is sufficient to guarantee $a^b \geq a^c$?
(The issue here is that B and D are both sufficient so the wording of B needs to be fixed)

A. $b \geq c$.

B. $a^b \geq a^c$.

C. $a^{b+1} \geq a^{c+1}$.

D. $a = 1$.

X2. A player wins one elementary game with probability $\frac{2}{3}$. To win the next-level competition, the player must win two games before losing two. The same recursive procedure defines successive competition levels.
(Not enough information for this one unfortunately)

X3. A point P lies on the circle $x^2 + y^2 = 1$. A real parameter m defines a line L, whose perpendicular distance from the origin is $2|m|/\sqrt{m^2+1}$.
(Not enough information for this one unfortunately, no equation of L etc.)

X4. Let $f(x) = ax^2 + bx + c$. The curve $y = g(x)$ is obtained from $y = f(x)$ by a vertical stretch of factor 2, and $y = h(x)$ by a translation six units to the right. It is given that

$$g(x) + h(x) = 30x^2 + 30x + 30.$$

(I assume they wanted you to find an expression in terms of a, b, c but I don't know what it is)

X5. A sequence (a_n) is described by

$$a_{n+1} = a_n^2 - k,$$

where the terms a_n are prime numbers for $n > 1$.
(The required statement, initial condition and any restrictions on k aren't known)

X6. Negate the statement: "For every rational number x, the integral of f is equal to $g(x)$." (Not sure what the original integral's limits were and what the differential variable was)